Revisiting Local Context for Long-Horizon Streaming 3D Reconstruction

AMAP CV Lab

Alibaba Group

Abstract

Streaming 3D reconstruction from extremely long videos requires estimating camera motion and scene geometry online under bounded memory and computation. Early methods use short temporal contexts or compact recurrent states but often degrade on long sequences. Recent methods address this limitation by augmenting short-range context with persistent or multi-level long-range memory. We pursue a different route to long-horizon streaming reconstruction: rather than maintaining long-range state, we formulate local predictions whose targets remain independent of sequence length. We present *ABot-Recon*, a simple streaming model that caches KV features from only the preceding 11 frames. It predicts a point map in the current camera coordinate system together with an adjacent-frame relative pose. These predictions remain equivariant under changes of reference frame, and global poses and geometry are recovered through sequential composition. To reduce accumulated drift, a lightweight temporal refiner improves relative rotations using recent visual and motion context, while a composition-aware pose loss supervises multi-step pose composition. Extensive evaluations on challenging long-sequence benchmarks demonstrate the superior long-horizon performance of our local-context approach, with *ABot-Recon* attaining an ATE of 4.35 m and an RPE-R of 0.12° on Oxford Spires.

Project Page: `https://example.com`
Code: `https://github.com/example/project`

1 Introduction

Recovering camera motion and dense 3D scene geometry from a video stream is a fundamental capability for robotics, autonomous driving, and embodied intelligence. Recent feed-forward reconstruction models [1–4] have shown that point maps and camera parameters can be inferred directly from unposed images using learned geometric priors. Most of these models, however, operate on a fixed collection of views through pairwise or global interactions. Ultra-long video streams change the nature of the problem: observations arrive causally, memory and per-frame computation must remain bounded, and the current view can be separated from early reference frames by thousands of time steps and substantial camera displacement.

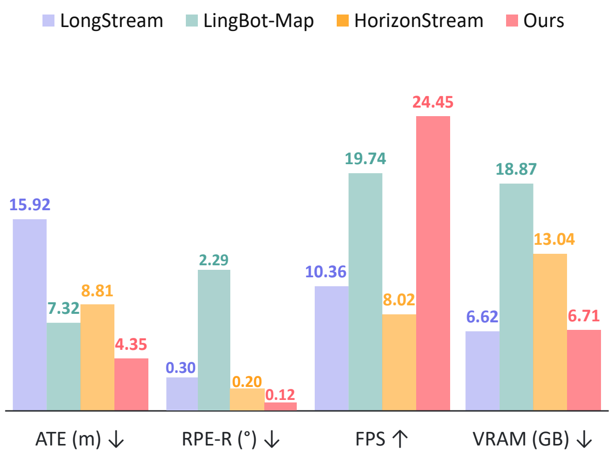


Figure 1. Pose accuracy on Oxford Spires and streaming inference efficiency on KITTI-02, measured by ATE, RPE$_r$, FPS, and peak GPU memory excluding input storage, on an NVIDIA H100 GPU.

Early streaming reconstruction models [5–10] adapt learned geometric priors to online inference through recurrent states, causal attention, or short temporal windows. These architectures process incoming observations efficiently, yet their pose and geometry estimates degrade when the deployment horizon extends far beyond the sequences seen during training. Long-horizon methods [11–15] expand the usable temporal range through gated recurrent states, explicit spatial or trajectory memories, cache compression, and specialized state-update rules. More recently, LingBot-Map [16]

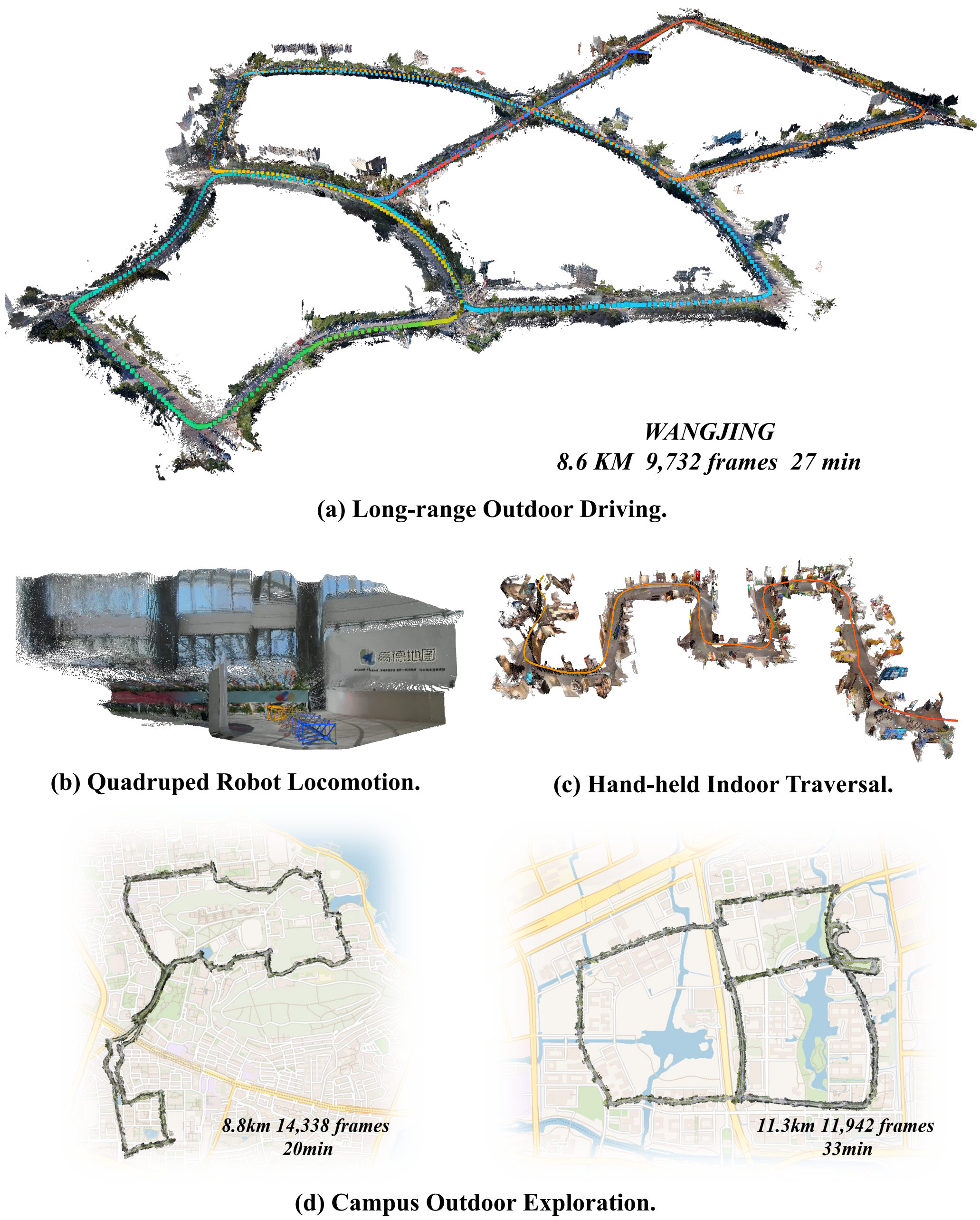


Figure 2. Visualization of reconstruction result of *ABot-Recon* with loop closure. The visualizations demonstrate the versatility of our method across diverse platforms, spanning large-scale outdoor driving and hand-held indoor exploration to quadruped robot locomotion.

and HorizonStream [17] have pushed streaming reconstruction beyond 10,000 frames by explicitly coordinating short-range observations with persistent long-range information. These advances demonstrate the power of long-range memory, while making the retention, selection, propagation, and fusion of historical context a central part of the architecture.

We take a different route: we keep the learned temporal context strictly local and make its predictions reliably composable over time. The key is to prevent the prediction target from expanding with the sequence. Rather than estimating every camera and point map in a unified coordinate system tied to a distant first frame, the model predicts geometry and motion within a fixed local range. It therefore solves the same local estimation problem at every time step, while the global trajectory and reconstruction are assembled incrementally from local measurements.

Based on this principle, we introduce *ABot-Recon*, a streamlined streaming reconstruction model whose learned components operate within a 12-frame temporal horizon. For each incoming frame, the model attends only to KV features cached from the preceding 11 frames and maintains no persistent learned long-range state. *ABot-Recon* predicts a point map P_i in the current camera coordinate system and an adjacent-frame transformation $T_{i-1\leftarrow i}$. Its local predictions transform consistently under changes of reference frame, avoiding dependence on a fixed global anchor. Global camera poses are recovered by chaining the adjacent transformations, and the local point maps are transformed accordingly into the global reconstruction. This design keeps model memory and per-frame computation independent of the total sequence length.

The central challenge is then to ensure that local pose estimates remain stable under repeated composition. We address it at both the estimation and supervision levels. A lightweight temporal rotation refiner conditions each relative rotation on recent motion evolution and dense visual evidence, correcting unreliable estimates before they enter the trajectory. A composition-aware pose objective supervises transformations over multiple temporal spans, optimizing local predictions according to their accumulated effect rather than treating each adjacent pair in isolation. Together, these components stabilize the pose chain without extending the model's temporal context. For sequences with repeated revisits, *ABot-Recon* can optionally incorporate a training-independent SLAM-style loop-closure backend at inference time to further improve reconstruction quality, without modifying the training procedure.

Extensive experiments show that *ABot-Recon* generalizes from short training sequences to ultra-long video streams and delivers superior camera tracking and 3D reconstruction performance on challenging long-sequence benchmarks, including KITTI [18] and Oxford Spires [19]. Without loop closure, *ABot-Recon* attains an ATE of 4.35 m and an RPE-R of 0.12° on Oxford Spires; the optional loop-closure backend further reduces the ATE to 3.99 m. These results demonstrate that a learned model with only local temporal context can recover accurate and locally stable trajectories over extended spatial and temporal horizons.

Our contributions are summarized as follows:

- We establish a local-context formulation for ultra-long streaming 3D reconstruction, keeping the reference frames and estimation ranges of local prediction targets fixed as the sequence grows.
- We introduce *ABot-Recon*, a streamlined model that uses only a 12-frame temporal horizon, consisting of the current frame and KV features from the preceding 11 frames, and recovers global poses and geometry from reference-frame-consistent local predictions.
- We stabilize long-horizon pose composition through lightweight temporal rotation refinement and composition-aware pose supervision, achieving superior performance on challenging long-sequence benchmarks without persistent learned long-range memory.

2 Related Work

2.1 Feed-Forward Visual Geometry Reconstruction

Classical structure-from-motion (SfM) and visual SLAM systems [20, 21] recover camera motion and scene structure through carefully engineered pipelines that combine correspondence estimation, geometric verification, triangulation, bundle adjustment, and loop closure. Recent feed-forward methods replace a substantial part of this pipeline with learned geometric priors. DUSt3R [1] introduced pointmap regression in a shared coordinate frame, enabling two-view reconstruction without known camera intrinsics or poses, while MASt3R [2] augmented this representation with dense local features for accurate image matching.

Subsequent methods extended this paradigm to broader reconstruction settings. MV-DUSt3R+ [22] and Fast3R [23] directly exchange information across multiple input images, while VGGT [3] jointly predicts cameras, depth, point maps, and tracks through global multi-view interaction. Pi3 [4] imposes permutation equivariance over the input views, FLARE [24] couples camera estimation with geometry and appearance prediction, PLANA3R [25] represents indoor scenes using metric planar primitives learned through planar splatting, and MonST3R [26] extends feed-forward reconstruction to dynamic videos. These models demonstrate that strong learned priors can replace much of the conventional reconstruction pipeline. However, they either process a bounded collection of views jointly or require pairwise or global alignment, and therefore do not by themselves provide causal inference with bounded cost over an indefinitely growing video stream.

2.2 Streaming Reconstruction and Temporal Memory

Streaming reconstruction methods process new observations incrementally by maintaining information from previously seen frames. Spann3R [5] stores past geometric features in an external spatial memory and predicts new point maps in a global coordinate system. CUT3R [6] instead maintains a compact recurrent state that is continuously updated by incoming observations. SLAM3R [10] reconstructs overlapping local clips and progressively registers their point maps without explicitly estimating camera parameters. A parallel line adopts causal Transformers: StreamVGGT [7] converts global multi-view attention into temporal causal attention with cached keys and values, and STream3R [8] formulates sequential pointmap prediction with a decoder-only causal Transformer. WinT3R [9] combines sliding-window interaction with a global camera-token pool, whereas Point3R [27] organizes historical information as an explicit spatial pointer memory. Together, these approaches establish effective architectures for online reconstruction, but their temporal state is still responsible for both local geometric estimation and the preservation of global consistency.

Several methods explicitly extend streaming reconstruction beyond short sequences by strengthening how temporal state is retained, selected, or updated. LONG3R [11] introduces gated 3D spatio-temporal memory and curriculum training over increasing sequence lengths. LongStream [12] replaces first-frame anchoring with keyframe-relative prediction and combines cache-consistent training with periodic cache refresh, enabling reconstruction over thousands of frames and kilometer-scale trajectories. STAC [13] compresses cached spatio-temporal features to control the cost of retaining a long history. TTT3R [14] improves recurrent-state retention through a training-free, confidence-adaptive update rule. Mem3R [15] further decouples camera tracking from geometric mapping, using a test-time-trained fast-weight memory for tracking and a fixed-size token state for mapping.

More recently, LingBot-Map [16] and HorizonStream [17] push streaming reconstruction into an ultra-long regime exceeding 10,000 frames, where current observations may be separated from early references by both vast temporal gaps and large camera displacement. LingBot-Map coordinates anchor context, a pose-reference window, and trajectory memory to preserve geometric information at multiple temporal scales. HorizonStream factorizes geometric propagation into short-range local attention and persistent linear attention, allowing evidence to propagate over long horizons with bounded memory. These works shift the focus from delaying degradation on longer clips to maintaining stable geometry throughout continuous, large-scale exploration. They also demonstrate the effectiveness of explicitly coordinating short- and long-range information. We study a different design point: our learned model retains only the KV cache of the most recent K frames and contains no persistent or hierarchical long-range memory. Instead of asking the network to preserve an increasingly long reconstruction history, we formulate its outputs as local geometric measurements whose prediction difficulty is independent of the elapsed sequence length.

2.3 Local-to-Global Pose and Trajectory Recovery

The separation between local motion estimation and global trajectory recovery is central to classical visual odometry and SLAM. Systems such as DROID-SLAM [28] estimate dense inter-frame constraints and optimize them over a graph, while ORB-SLAM3 [21] uses keyframes, bundle adjustment, and loop closure to correct accumulated error. Learned geometric priors have recently been incorporated into similar pipelines. MASt3R-SLAM [29] uses MASt3R pointmap matching for tracking, graph construction, loop closure, and global optimization. Window-based reconstruction systems instead form local submaps and align them over time: SLAM3R [10] learns progressive registration, VGGT-Long [30] combines chunked VGGT inference with overlap alignment and loop closure, and TALO [31] introduces long-term, high-degree-of-freedom submap alignment. Anchor3R [32] predicts current-centered relative-pose measurements inside a transient window and integrates them through an explicit motion graph. R3 [33] improves global pose estimation through relative pose prediction, but remains anchored to the first-frame reference frame. These works confirm the value of local-to-global decomposition, but global consistency is primarily obtained through an additional registration or graph-optimization stage.

Our formulation brings this principle into the learned streaming predictor itself. Rather than regressing every camera in a coordinate frame permanently tied to the beginning of the sequence, the model predicts poses over bounded local prefixes and adopts a reference-frame-equivariant architecture. A pose-chain loss supervises compositions over multiple temporal spans, explicitly teaching locally estimated transformations to form stable long trajectories. Rotation refinement likewise uses only recent visual evidence and motion context, preserving the bounded-memory

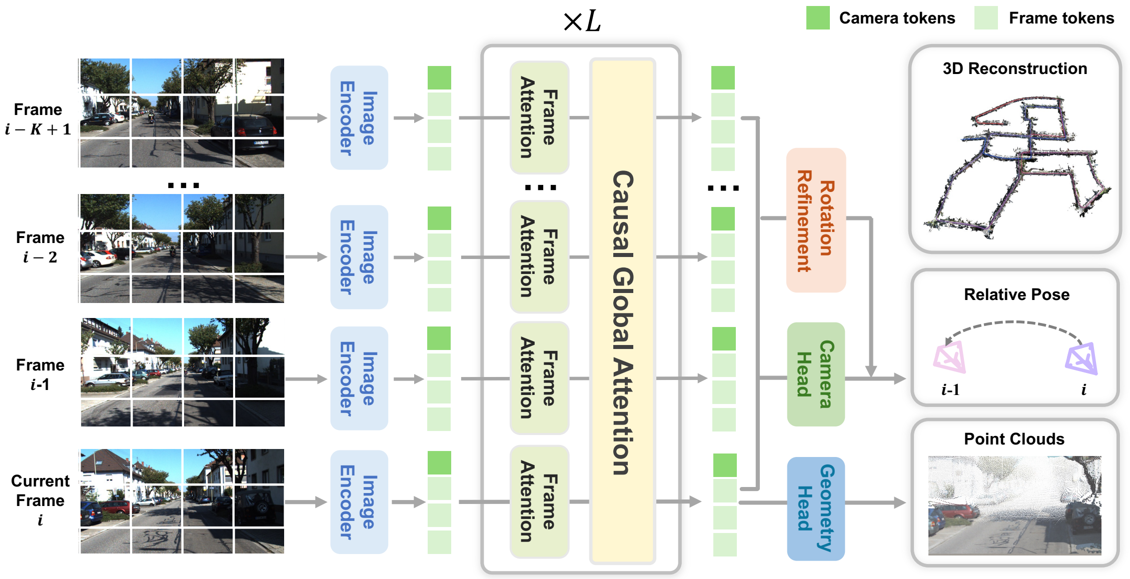


Figure 3. Pipeline of our method. For each incoming frame I, the model predicts within a fixed K-frame causal context. A shared image encoder followed by alternating intra-frame and windowed causal cross-frame attention predicts dense frame tokens and compact camera tokens. The geometry head predicts a local point map P_i in the current camera coordinate system, while adjacent camera-token descriptors are used to predict the relative transformation $T_{i-1\leftarrow i}$. A lightweight motion-visual rotation refiner combines the motion and visual evidence to estimate a residual rotation. The refined adjacent transformations are composed online to recover the global camera trajectory.

design. A conventional loop-closure backend can optionally refine the final trajectory at inference time, but it is independent of model training and is not required by the feed-forward reconstruction architecture. The resulting distinction is that long-horizon capability is learned through local prediction and reliable composition, rather than through learned access to long-range memory.

3 Method

3.1 Overview

Our key design principle is to formulate long-horizon streaming reconstruction as repeated local geometric estimation, thereby enabling a simple and scalable architecture. For each incoming frame I_i, *ABot-Recon* predicts a point map P_i in the current camera coordinate system and a relative pose $T_{i-1\leftarrow i}$ to the previous frame. The model operates with a fixed K-frame temporal context, set to 12 frames in our implementation, and maintains no persistent learned long-range state. Global camera poses and scene geometry are recovered by composing these local predictions over time, keeping model memory and per-frame computation independent of sequence length.

Section 3.2 describes the local prediction formulation and windowed causal attention. Section 3.3 introduces a lightweight temporal refiner that improves relative rotations using recent motion and visual context. Section 3.4 presents a composition-aware objective that supervises pose chains over multiple temporal spans.

3.2 Local Geometry Estimation

For each incoming frame I_i, our model predicts a point map P_i in the current-frame coordinate system and the relative pose $T_{i-1\leftarrow i}$ of the current frame I_i with respect to the previous frame I_{i-1}. Our model follows a recent causal-attention Transformer architecture:

$$\begin{aligned} F_i &= \mathrm{Encoder}(I_i), \\ (G_i, C_i, \mathcal{M}_i) &= \mathrm{Decoder}([F_i, C], \mathcal{M}_{i-1}), \\ P_i &= \mathrm{Head}_{\mathrm{point}}(G_i). \end{aligned} \tag{1}$$

where C denotes a set of learnable camera tokens, $C_i = \{c_i^\ell\}_{\ell=1}^L$ denotes their decoded representations for frame I_i and $\mathcal{M}_i$ donated the memory caches.

The relative pose is predicted directly from the camera-token representations of two adjacent

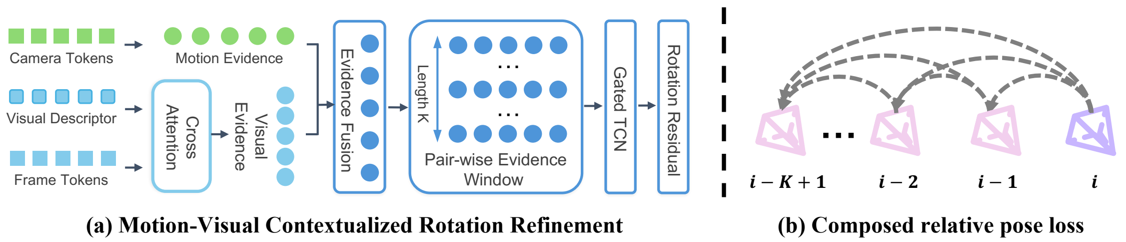


Figure 4. Motion-visual rotation refinement and composition-aware pose supervision. (a) For each adjacent frame pair, motion evidence are encoded from camera tokens, while the visual descriptors queries the corresponding dense frame tokens to form the visual evidence. Later, the two evidences are fused into a unified pairwise representation. A causal window is then aggregated by a lightweight gated TCN to predict a rotation residual, which refines the relative rotation. **(b)** Rather than supervising adjacent transformations independently, we compose the refined relative poses over multiple temporal gaps and compare each composed transformation with its corresponding ground truth, which directly exposes accumulated pose-chain errors during training.

frames. We first aggregate the camera tokens into a frame-level pose descriptor,

$$z_i = \frac{1}{L} \sum_{\ell=1}^{L} \mathrm{MLP}_{\mathrm{desc}} \left(c_i^{\ell} \right), \tag{2}$$

and construct a pairwise descriptor [34]

$$q_i = \mathcal{R}(z_{i-1}, z_i), \qquad \mathcal{R}(x, y) = [x, y, y - x, x \odot y], \tag{3}$$

The adjacent-frame relative pose is then obtained as

$$T_{i-1 \leftarrow i} = \mathrm{Head}_{\mathrm{pose}}(q_{i-1 \leftarrow i}) \tag{4}$$

The relative pose between any two frames I_i and I_j with $i < j$ can be recovered by chaining the predicted adjacent-frame transformations:

$$T_{i \leftarrow j} = T_{i \leftarrow i+1} T_{i+1 \leftarrow i+2} \cdots T_{j-1 \leftarrow j} = \prod_{k=i+1}^{j} T_{k-1 \leftarrow k}. \tag{5}$$

Since geometric prediction is performed locally, we further restrict the temporal context to a fixed-size window. Specifically, we replace full-history causal temporal attention with *windowed causal temporal attention*, where frame I_i attends only to the cached keys and values of the most recent $K-1$ frames:

$$\mathcal{M}_{j-1}^{(K)} = \{\mathrm{KV}_i\}_{i=\max(0, j-K+1)}^{j-1}. \tag{6}$$

Cached states are discarded once they fall outside the window, resulting in $O(K)$ temporal memory and $O(NK)$ temporal-attention computation over a sequence of N frames. For a fixed window size K, inference therefore requires constant memory while computation grows linearly with the sequence length. As this design only restricts the attention context, it remains easy to train and fully compatible with standard attention implementations, yielding a simple and scalable streaming backbone.

3.3 Motion-Visual Contextualized Rotation Refinement

Although adjacent-frame relative pose prediction avoids direct extrapolation in the global coordinate system, it can still be unreliable under rapid camera motion or challenging visual conditions. Moreover, predicting adjacent-frame relative pose independently ignores temporal motion consistency, causing local errors to accumulate through recursive pose composition and leading to long-term drift. To mitigate this issue, we introduce a Motion-Visual Contextualized refinement module that leverages recent motion history to refine unreliable relative rotations before trajectory composition.

For each adjacent frame pair, we construct a unified motion–visual evidence representation by integrating a compact inter-frame motion feature with spatially resolved visual evidence. Specifically, the inter-frame motion feature f_m is obtained by encoding the predicted pairwise motion descriptor q_i:

$$f_m = \phi_m(q_i). \tag{7}$$

To complement the global motion evidence, we extract pair-specific visual evidence in a coarse-to-fine manner. We first average the dense patch tokens of each frame to obtain compact visual descriptors $\bar{G}_{i-1}$ and $\bar{G}_t$. Their inter-frame relation is then used as a query to retrieve fine-grained information from the original dense frame tokens:

$$f_v = \mathrm{CrossAttn}\big(\phi_v\big(\mathcal{R}(\bar{G}_{i-1}, \bar{G}_i)\big), [G_{t-1}; G_t]\big)\,. \tag{8}$$

The relational query provides a coarse characterization of the visual variation between adjacent frames, while cross-attention [35] retrieves the corresponding spatially resolved evidence from the dense features, yielding the pair-conditioned visual representation v_t.

Finally, the motion and visual features are fused to yield a compact pairwise evidence that jointly captures the predicted camera motion and its supporting visual context as

$$f_c = \phi_{\mathrm{fuse}}([f_m, f_v])\,, \tag{9}$$

We then model the current pairwise feature together with its $K-1$ preceding features, forming a causal temporal window of length K. And a lightweight gated temporal convolutional network (TCN) [36] is introduced to aggregate the recent motion-visual context:

$$\begin{aligned} \mathcal{W}_i &= [f_{i-K+1}, \ldots, f_i], \\ \delta\omega_t &= \phi_o(\mathcal{T}_h(\mathcal{W}_i) \odot \sigma(\mathcal{T}_g(\mathcal{W}_i)))\,. \end{aligned} \tag{10}$$

where $\mathcal{T}_h$ and $\mathcal{T}_g$ denote two causal TCN branches for temporal feature aggregation and adaptive gating, respectively. By contextualizing the current pairwise estimate with recent motion-visual dynamics, the predicted residual can compensate for locally unreliable rotation estimates while preserving online causality.

Since adjacent-frame translation is generally more constrained and stable than rotation, we preserve the predicted translation and apply temporal refinement only to the rotational component. The predicted axis–angle residual $\delta\omega_t \in \mathbb{R}^3$ is mapped to SO(3) through the exponential map and composed with the initial relative rotation:

$$\widehat{R}_{i-1\leftarrow i} = \widetilde{R}_{i-1\leftarrow}\, \mathrm{Exp}([\delta\omega_t]_\times). \tag{11}$$

This residual formulation allows the temporal motion-visual context to selectively correct unreliable rotation estimates while retaining the original pairwise prediction as a reference. The refined relative poses are subsequently used for trajectory composition.

3.4 Composition-aware Loss Functions

As the refined relative poses are recursively composed to recover the camera trajectory, local prediction errors may accumulate over time. We therefore supervise composed relative poses over multiple temporal gaps rather than only adjacent-frame transformations, explicitly constraining the accumulated trajectory error.

For any frame pair (i, j) with $0 \le i < j < N$, the predicted transformation from frame j to frame i is obtained by composing the refined adjacent transformations. We supervise all frame pairs within a maximum temporal gap D:

$$\mathcal{P} = \{(i, j) \mid 0 \le i < j < N,\ j - i \le D\}\,. \tag{12}$$

To emphasize longer composition chains, each pair is assigned a normalized gap-dependent weight

$$\alpha_{ij} = \frac{(j-i)^\gamma}{\frac{1}{|\mathcal{P}|}\sum_{(m,n)\in\mathcal{P}}(n-m)^\gamma}, \qquad 0 < \gamma < 1. \tag{13}$$

Following Pi3 [4], we use the same translation and rotation losses for each composed relative pose:

$$\mathcal{L}_{\mathrm{pose}} = \frac{1}{|\mathcal{P}|}\sum_{(i,j)\in\mathcal{P}} \Big(\lambda_{\mathrm{trans}}\ell_{\mathrm{trans}}^{(i,j)} + \alpha_{ij}\lambda_{\mathrm{rot}}\ell_{\mathrm{rot}}^{(i,j)}\Big)\,. \tag{14}$$

Since the loss is evaluated on composed transformations, it directly supervises the accumulated trajectory error over different temporal spans.

Table 1. Training-data composition. Sampling ratios are obtained by normalizing the dataset weights used by the training sampler. Internally collected data are grouped by provenance as *Internal Synthetic* and *Internal Real*.

Synthetic data		Real-world data	
Dataset	Ratio (%)	Dataset	Ratio (%)
TartanAir-v2 [37]	8.96	DL3DV [38]	8.96
TartanAir [37]	6.72	Waymo [39]	5.38
OmniWorld-Game [40]	8.96	ARKitScenes [41]	1.34
TartanGround [42]	6.72	ScanNet++ [43]	3.81
VirtualKITTI2 [44]	6.27	WildRGBD [45]	2.91
HyperSim [46]	6.27	ScanNet [47]	2.46
PointOdyssey [48]	0.90	MapFree [49]	1.12
Unreal4K [50]	1.34	ARKitScenes HR [51]	2.46
Spring [52]	0.22	UASOL [53]	0.45
MVS-Synth [54]	0.67	DDAD [55]	1.34
Dynamic Replica [56]	0.45	KITTI-360 [57]	2.02
ASE [58]	2.69	Holo360D [59]	1.34
MidAir [60]	2.69	Holo360D_val [59]	0.67
MatrixCity [61]	1.34	Internal Real	3.67
AirZoo [62]	1.34		
BlendedMVS [63]	2.46		
Internal Synthetic	4.03		
Synthetic total	**62.05**	**Real-world total**	**37.95**

Table 2. Evaluation datasets and statistics.

Dataset	Task	Frames/sequence	#Seq.	Infer stride
KITTI [18]	Camera pose	271–4,661	11	1
VBR [64]	Camera pose	8,815–18,846	7	1
Oxford Spires [19]	Pose / Reconstruction	3,821–3,840	10	1
7Scenes [65]	Reconstruction	500–1,000	7	1
TUM-Dynamic [66]	Reconstruction	707–1,261	8	1

We additionally introduce a residual regularization term $\mathcal{L}_{\text{smooth}}$ that penalizes the magnitude and temporal variation of the predicted rotation residuals, encouraging small and temporally consistent refinements.

For local geometry, we retain the point-map loss $\mathcal{L}_{\text{pts}}$ and surface-normal loss $\mathcal{L}_{\text{normal}}$ [4], which supervise local 3D geometry and surface orientation, respectively.

The overall training objective is

$$\mathcal{L} = \lambda_{\text{pose}}\mathcal{L}_{\text{pose}} + \lambda_{\text{smooth}}\mathcal{L}_{\text{smooth}} + \lambda_{\text{pts}}\mathcal{L}_{\text{pts}} + \lambda_{\text{normal}}\mathcal{L}_{\text{normal}}, \tag{15}$$

where the λ coefficients balance the corresponding supervision terms.

4 Experiments

4.1 Experimental Setup

Training data. We train on a diverse mixture of 32 synthetic and real-world datasets covering indoor scenes, outdoor environments, autonomous driving, handheld capture, and aerial trajectories. As summarized in Table 1, synthetic and real-world data account for 62.05% and 37.95% of the sampling distribution, respectively. We assign larger sampling probabilities to datasets containing long, temporally coherent trajectories, while sampling from the remaining datasets to preserve diversity in scene appearance, geometry, and camera motion.

Data processing and augmentation. We convert all camera poses to a unified camera-to-world convention, normalize dataset-specific depth and translation units, and discard samples with invalid poses, missing frames, or corrupted geometry. For video datasets, frames are sampled in their original temporal order using dataset-specific intervals. For unordered multiview datasets, we construct pseudo-sequences from a camera pose graph that favors locally coherent neighboring views We apply focal-length and crop perturbations and preserve the source field of view by resizing to the target width followed by center cropping or mean-color padding. For datasets with notably off-center cameras, we retain the original non-central principal point with probability 0.9 instead of always applying principal-point-centered cropping. Photometric augmentation includes brightness, contrast, saturation, hue, and gamma jitter, as well as JPEG degradation and image blur. Dataset-specific temporal intervals and augmentation parameters are provided in the supplementary material.

Training stages. Training consists of two main stages with a progressively extended temporal horizon, followed by a short confidence-calibration phase. We initialize Stage I from the publicly released π^3 weight [4], thereby retaining its pretrained image encoding and geometric reconstruction capability. We replace the original absolute camera-pose output with our adjacent-frame relative pose parameterization and adapt the model to causal streaming inference. Stage I uses 32-frame clips and is trained for 48K iterations with a batch size of 48 on 48 NVIDIA H20 GPUs. This stage establishes stable local geometry and adjacent motion estimation.

In Stage II, we initialize from the Stage I checkpoint and extend the training clips from 32 to 128 frames, while keeping the local prediction window fixed at 12 frames. Longer clips provide denser temporally continuous pose supervision and expose the model to more complete pose-composition chains within each training sample. We also enable the motion–visual contextualized rotation refinement module, which refines local rotation estimates before they are composed into the trajectory. Stage II is trained for another 38k iterations with a batch size of 64 on 64 AMD MI308 GPUs.

Finally, we introduce the confidence prediction branch and fine-tune it for an additional 4k iterations. During this phase, the reconstruction and pose estimation networks are kept fixed, and only the confidence decoder and prediction head are optimized. This confidence-only fine-tuning allows the model to estimate the reliability of its dense predictions without altering the geometry and camera trajectories learned in the preceding stages.

Implementation details. We use the AdamW optimizer [67] throughout training. In Stage I, the learning rate reaches a peak of 1×10^{-4} and is gradually annealed to 1×10^{-6}. In Stage II, we adopt differential learning rates: the newly introduced rotation refinement module is optimized with a peak learning rate of 5×10^{-5}, while the remaining model parameters are fine-tuned with a smaller peak learning rate of 2×10^{-5}. This setting allows the new module to adapt effectively while limiting disruption to the local pose and geometry representations learned in Stage I. During the final confidence-calibration phase, only the confidence decoder and prediction head are optimized, using a peak learning rate of 2×10^{-5}.

We maintain an exponential moving average (EMA) [68] with a decay coefficient of 0.999 throughout training. During the confidence-calibration phase, EMA is applied only to the trainable confidence parameters. Parameter averaging smooths the optimization trajectory and reduces sensitivity to transient noisy or unusually large updates, which is particularly useful for our heterogeneous training mixture. We use a per-GPU batch size of one sequence, bfloat16 mixed precision, and gradient clipping at 1.0.

Input images are resized to a width of 504 pixels while preserving their aspect ratio. The resulting images are then center-cropped or mean-padded vertically to obtain an input resolution of 504×280. The same causal inference rule is applied to all test sequences.

Inference mode. At inference time, the model processes the input stream causally, without resetting its streaming state. The backbone maintains only the local causal KV cache described above, which is implemented using the paged KV-cache operators in FlashInfer [69] for efficient streaming attention. Each incoming frame is processed once to predict its local geometry and adjacent-frame relative pose, after which the global camera trajectory and point cloud are recovered incrementally through pose composition.

Our model can be used directly without any additional long-range memory or online optimization. When stronger global consistency is desired, we optionally attach a training-independent SLAM-style loop-closure backend [17, 30] that refines the accumulated trajectory while leaving the underlying streaming prediction process unchanged. During loop closure, revisited frame pairs are retrieved from historical observations via FAISS-based search [70] over DINOv2-SALAD descriptors [71]. Local windows around each selected frame pair are then combined and re-fed into the model to predict their relative pose. The resulting predictions are converted into loop-closure constraints and incorporated into sparse pose graph optimization to refine the global trajectory.

Evaluation datasets. We evaluate our method on two complementary tasks: camera pose estimation and dense 3D reconstruction. For camera pose estimation, we use KITTI [18], Oxford Spires [19], and VBR [64]. These benchmarks cover kilometer-scale driving sequences as well as handheld and vehicle-mounted capture. For dense 3D reconstruction, we report results on 7Scenes [65] , TUM-Dynamic [66], and Oxford Spires [19], covering indoor and outdoor environments as well as static and dynamic scenes. All methods process every evaluated sequence once in temporal order with a stride of 1. For 7Scenes, we select `seq-01` from each of its seven scenes; for all

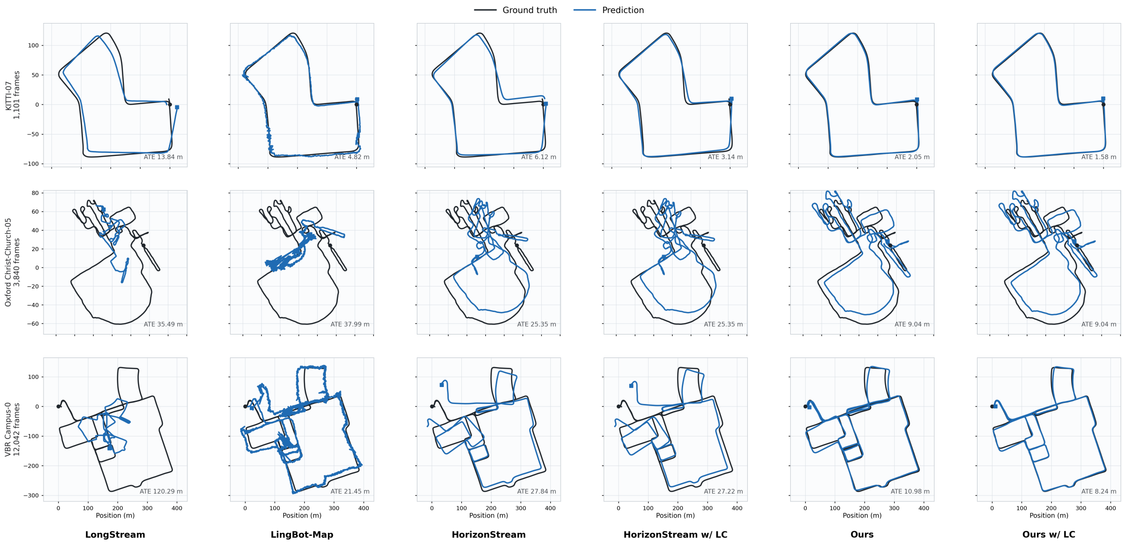


Figure 5. Qualitative camera pose comparison on representative sequences from KITTI, Oxford Spires, and VBR. Predicted trajectories (blue) are aligned to the ground truth (black) using Umeyama Sim(3) alignment, with the corresponding ATE reported in the lower-right corner. The selected sequences cover different trajectory lengths, scene scales, and motion patterns. Our method preserves the overall trajectory geometry consistently across the three benchmarks, while the optional loop-closure backend further reduces accumulated global drift.

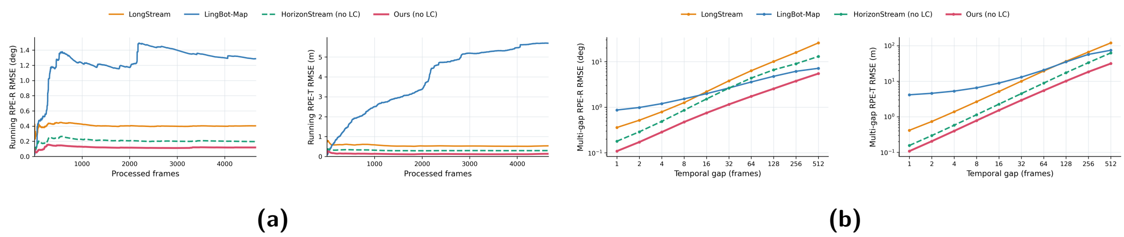


Figure 6. Long-horizon pose stability on KITTI. (a) Running RMSE of adjacent-frame RPE_r and RPE_t on the KITTI-02 sequence as the number of processed frames increases. (b) Multi-gap RPE_r and RPE_t averaged over all KITTI sequences as the temporal gap increases. All methods are evaluated causally without loop closure.

other benchmarks, we use all sequences in the corresponding evaluation set. Dataset statistics are summarized in Table 2.

Baseline methods. We compare against two groups of methods. Streaming reconstruction baselines include LingBot-Map [16], HorizonStream [17], LongStream [12], InfiniteVGGT [72], OVGGT [73], CUT3R [6], TTT3R [14], and Stream3R-w [8]. We additionally compare camera trajectories with optimization-based methods including VGGT-SLAM [74], MASt3R-SLAM [29], VGGT-Long [30], SLAMFormer-∞ [75], Droid-SLAM [28], DPV-SLAM [76], DPVO [77], and Droid-W [78]. Droid-SLAM, DPV-SLAM, DPVO, and Droid-W use camera intrinsics, whereas our method estimates camera motion directly from the RGB stream without requiring test-time intrinsics. Since SLAMFormer-∞ is not publicly available, we quote its KITTI ATE directly from the paper and all other baselines are evaluated using their publicly released official implementations and checkpoints under our unified protocol. CUT3R and TTT3R process each complete sequence with and without state reset, and Stream3R-w uses its official window mode with a window size of 5.

Evaluation protocol and metrics. All methods process each test sequence in temporal order and all benchmarks are evaluated with a temporal stride of 1. For camera pose evaluation, each predicted trajectory is aligned to the ground truth using Umeyama Sim(3) alignment. We report the RMSE of Absolute Trajectory Error (ATE), relative translational pose error (RPE_t), and relative rotational pose error (RPE_r). For dense reconstruction, predicted point maps are first resampled to the ground-truth evaluation grid using nearest-neighbor interpolation. Following LingBot-Map [16], the predicted and ground-truth point clouds are aligned using Umeyama registration, voxelized, and refined with point-to-point ICP. We use voxel sizes of 4/512 m for 7Scenes and TUM-Dynamic

Table 3. Camera pose estimation on KITTI. All metrics are lower better. Within streaming methods, best and second-best results are marked in blue and underlined, respectively. * denotes methods that require camera intrinsics, and ‡ denotes results quoted from the corresponding paper rather than reproduced under our evaluation pipeline. w/ LC indicates results with loop closure enabled.

		Per-sequence ATE (m) ↓											Sequence averages		
		00 4541f 3.70 km	01 1101f 2.50 km	02 4661f 5.10 km	03 801f 0.60 km	04 271f 0.40 km	05 2761f 2.20 km	06 1101f 1.20 km	07 1101f 0.70 km	08 4071f 3.20 km	09 1591f 1.70 km	10 1201f 0.90 km	Avg. ATE (m)↓	Avg. RPE_r (°)↓	Avg. RPE_t (m)↓
Group	Method														
Opt.-based	VGGT-SLAM	27.99	214.23	260.88	53.50	26.02	99.61	23.62	64.52	243.87	169.10	80.13	114.86	1.57	2.73
	MASt3R-SLAM	190.12	507.15	232.17	169.82	88.98	126.37	133.33	62.92	199.70	178.03	151.69	185.48	0.98	2.58
	VGGT-Long	8.63	58.37	52.79	8.83	4.23	9.83	4.65	2.67	73.12	31.84	27.67	25.69	0.26	0.42
	SLAMFormer-∞‡	15.39	96.58	28.81	4.97	4.16	11.36	13.54	8.53	37.00	18.85	13.93	23.01	–	–
	Droid-SLAM*	131.25	267.79	217.63	3.40	1.47	80.41	64.20	18.19	167.80	90.05	22.74	96.81	0.23	0.60
	DPV-SLAM*	113.19	16.76	113.06	2.44	0.99	59.42	53.32	18.90	110.36	74.60	13.74	52.43	0.05	0.27
	DPVO*	113.17	16.56	113.37	2.45	0.98	59.59	55.63	19.40	110.64	74.44	13.80	52.73	0.05	0.27
	Droid-W*	48.26	207.57	52.80	1.76	4.65	25.16	5.34	3.35	29.05	35.56	8.01	38.32	0.05	0.36
Streaming	InfiniteVGGT	186.48	624.69	289.03	168.42	73.88	143.81	116.65	85.53	225.14	214.28	156.87	207.71	9.44	19.21
	OVGGT	185.65	716.66	304.04	155.78	65.01	155.48	108.51	85.92	246.97	215.64	178.67	219.85	11.02	18.02
	Stream3R-w	188.66	695.02	301.28	161.68	98.99	159.85	122.82	87.19	264.61	219.90	194.32	226.76	7.03	11.52
	CUT3R	185.21	654.37	300.51	157.95	23.72	153.77	133.09	68.57	230.95	206.95	184.34	209.04	2.94	2.79
	CUT3R w/ reset	190.70	91.56	269.58	20.95	5.97	104.88	72.01	27.44	164.69	93.11	41.53	98.40	0.55	0.65
	TTT3R	165.12	524.16	272.38	105.06	12.32	146.24	131.02	61.75	262.88	187.14	124.13	181.11	3.85	3.49
	TTT3R w/ reset	118.19	102.98	242.05	14.37	3.52	31.62	37.66	11.99	81.52	79.10	29.81	68.44	0.49	0.60
	LongStream	90.02	63.46	236.42	5.04	2.57	80.62	12.76	13.84	89.61	95.07	25.67	65.01	0.35	0.33
	LingBot-Map	33.57	151.48	66.02	**2.72**	2.09	12.79	6.06	4.82	**13.36**	19.50	**8.01**	29.13	0.61	3.04
	HorizonStream	29.67	24.15	97.24	5.53	**0.72**	13.94	**5.46**	6.12	22.63	29.79	12.35	22.51	0.17	0.11
	HorizonStream w/ LC	15.45	24.15	71.01	5.53	**0.72**	7.84	7.54	3.14	22.63	27.45	12.35	17.98	0.30	**0.09**
	Ours	22.36	**21.50**	69.98	4.36	2.91	13.53	12.05	2.05	22.36	**14.06**	16.11	18.30	**0.10**	0.10
	Ours w/ LC	**10.78**	**21.50**	**63.45**	4.36	2.91	**7.36**	13.25	**1.58**	22.36	16.25	16.11	**16.36**	**0.10**	0.10

and 0.05 m for Oxford Spires, and report Chamfer Distance (CD) and F1 at a distance threshold of 0.25 m (7Scenes and TUM-Dynamic) and 4 m (Oxford Spires). For Oxford Spires, inference is performed on every frame, while point-cloud evaluation uses every 10th frame to keep evaluation tractable. Lower is better for all metrics except F1.

4.2 Camera Pose Estimation

Overall results. Tables 3, 4, and 5 report camera pose estimation results on KITTI, Oxford Spires, and VBR, respectively. Across the three benchmarks, our method achieves strong long-horizon trajectory accuracy using only a compact local temporal context, without maintaining a persistent global reference or long-range temporal memory.

On KITTI, our method achieves the lowest average ATE among the compared methods without and with loop closure, respectively, while also obtaining strong relative-pose accuracy. On Oxford Spires, our method achieves an average ATE of 4.35 m without loop closure, together with the best RPE_r and RPE_t among the streaming methods. The optional loop-closure backend further reduces the average ATE to 3.99 m. Although some optimization-based methods (DPV-SLAM, DPVO, and Droid-W) obtain lower ATE on this benchmark, these methods require camera intrinsics, whereas ours operates solely on the RGB stream.

On VBR, which consists of substantially longer sequences under diverse capture conditions, our method remains competitive with HorizonStream and LingBot-Map in global trajectory accuracy without loop closure, while achieving relative-pose accuracy comparable to HorizonStream. With the optional loop-closure backend, our method further improves global trajectory accuracy, achieving the lowest average ATE among the compared methods.

Qualitative trajectory comparison. Figure 5 provides a complementary view of trajectory quality beyond aggregate ATE. Across the three benchmarks, our method preserves the overall trajectory geometry while producing smooth local motion. On particularly long trajectories, our relative-pose formulation exhibits less local trajectory jitter than the anchor-based LingBot-Map, while achieving a level of local stability comparable to HorizonStream despite using only a local context.

Long-horizon error stability. To quantitatively characterize the local trajectory stability observed above, we first examine how adjacent-frame relative-pose errors evolve as the processing horizon increases. Figure 6 (a) reports the running RMSE of RPE_r and RPE_t on the 4,661-frame KITTI-02 sequence, computed over all adjacent frame pairs observed up to each processed frame. Our method maintains the lowest running error throughout the sequence, with no progressive degradation as more frames are processed. This behavior is consistent with the length-independent nature of our relative-pose formulation: the regression target remains local regardless of the total stream length.

Stable adjacent-frame predictions, however, do not necessarily guarantee accurate long-term pose

Table 4. Camera pose estimation on Oxford Spires. Within streaming methods, best and second-best results are marked in blue and underlined, respectively. * denotes methods that require camera intrinsics.

Group	Method	Keble-2 3840f 0.20 km	Keble-3 3840f 0.18 km	Keble-4 3840f 0.23 km	Keble-5 3840f 0.21 km	Observ.-1 3840f 0.27 km	Observ.-2 3839f 0.28 km	Christ-2 3821f 0.25 km	Christ-3 3840f 0.18 km	Christ-5 3840f 0.80 km	Bodleian-2 3840f 0.55 km	Avg. ATE (m)↓	Avg. RPE$_r$ (°)↓	Avg. RPE$_t$ (m)↓
		Per-sequence ATE (m) ↓										Sequence averages		
Opt.-based	VGGT-SLAM	2.64	3.78	3.89	1.86	5.74	3.39	7.41	1.02	12.94	18.83	6.15	0.39	0.05
	MASt3R-SLAM	7.06	18.03	11.27	14.64	10.26	10.30	2.90	6.95	19.85	19.99	12.12	0.36	0.12
	VGGT-Long	9.07	9.37	11.77	3.96	6.03	4.05	2.41	0.48	14.38	12.59	7.41	0.29	0.08
	Droid-SLAM*	16.79	15.99	29.40	21.00	0.87	1.08	1.31	1.15	7.58	1.82	9.70	0.12	0.03
	DPV-SLAM*	0.20	1.31	0.03	0.30	0.16	0.45	0.15	0.04	25.77	1.08	2.95	0.15	0.03
	DPVO*	0.12	0.84	0.05	0.29	0.16	0.51	0.15	0.35	12.04	1.08	1.56	0.16	0.01
	Droid-W*	2.45	0.99	2.64	4.27	1.07	1.06	1.63	0.75	1.56	6.28	2.27	0.08	0.01
Streaming	InfiniteVGGT	25.02	20.39	25.09	24.15	29.10	28.14	33.02	12.95	40.39	79.32	31.76	5.30	2.70
	OVGGT	32.97	23.01	28.53	23.23	28.51	27.75	30.01	13.09	42.35	82.41	33.19	7.83	3.61
	Stream3R-w	31.28	22.39	31.07	24.63	28.38	27.18	38.12	12.99	42.39	82.66	34.11	9.05	2.67
	CUT3R	31.68	21.67	22.76	25.07	26.77	29.41	33.93	13.06	43.66	79.93	32.79	1.78	0.25
	CUT3R w/ reset	13.10	11.55	15.76	8.98	13.58	15.60	12.36	5.71	31.31	19.42	14.74	0.66	0.05
	TTT3R	24.66	22.56	27.33	18.67	26.94	28.18	19.83	9.68	26.76	55.01	25.96	3.19	0.31
	TTT3R w/ reset	8.27	7.26	8.77	7.55	9.43	10.30	4.56	2.11	21.75	14.00	9.40	0.29	0.04
	LongStream	11.57	9.44	16.88	10.67	12.43	14.34	17.93	6.42	35.49	24.02	15.92	0.30	0.06
	LingBot-Map	**2.71**	3.04	2.04	**3.23**	5.38	4.43	**2.21**	0.74	37.99	11.43	7.32	2.29	0.53
	HorizonStream	5.79	6.47	7.36	6.28	6.33	5.30	10.69	3.11	25.35	11.41	8.81	0.20	0.03
	HorizonStream w/ LC	5.79	6.47	**1.25**	6.24	6.33	5.30	10.69	**0.47**	25.35	11.41	7.93	0.20	0.03
	Ours	4.23	1.70	5.10	4.47	**4.58**	**4.41**	3.10	1.16	**9.04**	**5.76**	4.35	**0.12**	**0.02**
	Ours w/ LC	4.23	**1.69**	1.99	4.47	**4.58**	**4.41**	3.10	0.66	**9.04**	**5.76**	**3.99**	**0.12**	**0.02**

composition, as local errors can accumulate through repeated composition. We therefore evaluate multi-gap RPE over all KITTI sequences in Figure 6 (b). Our method achieves the lowest average rotational and translation error at every evaluated gap. The slower error growth at long gaps is consistent with the role of composed-pose supervision, which directly constrains the multi-step predictions used to assemble the streaming trajectory.

Runtime and memory efficiency. We benchmark forward throughput and peak GPU memory, excluding input storage, on a single NVIDIA H100 GPU. All methods use frame-by-frame streaming inference, while LongStream* and HorizonStream* denote their default segment- and chunk-based modes. Our method achieves 25.11 FPS with 6.34 GB memory, offering a favorable balance between streaming efficiency and long-horizon pose accuracy. HorizonStream* is faster through temporal batching but consumes over four times more memory, while CUT3R and TTT3R exhibit larger long-sequence drift.

4.3 Dense 3D Reconstruction

Table 7 reports dense reconstruction results on 7Scenes, TUM-Dynamic, and Oxford Spires, covering compact indoor scenes, dynamic RGB-D sequences, and larger landmark-scale environments. Across these different settings, our method achieves competitive reconstruction quality.

On 7Scenes and TUM-Dynamic, our method remains competitive with strong streaming baselines across both CD and F1. Although these sequences can span hundreds or thousands of frames, they are captured within relatively compact indoor environments, where camera observations remain spatially confined and exhibit substantial visual overlap over time. Persistent geometric context can therefore remain relevant throughout the stream. Nevertheless, our method retains comparable reconstruction quality across both static and dynamic indoor scenes without maintaining a persistent global anchor or long-range geometric memory.

On Oxford Spires, the sequences cover substantially larger landmark-scale environments than the compact indoor benchmarks. Our method achieves the lowest CD and the highest F1 among the compared methods. This result indicates that restricting geometric prediction to local temporal context remains effective when reconstruction spans a much larger spatial extent.

Figure 7 further compares the reconstructed geometry on representative sequences of different lengths and scene scales. Across KITTI-02, Oxford Obs.-Q1, and 7Scenes Stairs-01, all three methods recover the main scene structure, while our local formulation maintains coherent geometry from compact indoor scenes to substantially longer driving and landmark-scale sequences. The zoomed regions on KITTI and Oxford further show that local geometric structure remains well aligned within the global reconstruction.

4.4 Ablation Studies

We conduct controlled ablation experiments on the KITTI, Oxford, and VBR datasets which cover different motion patterns, scene structures, and acquisition conditions, to evaluate the contributions

Table 5. Camera pose estimation on VBR. Within streaming methods, best and second-best results are marked in blue and underlined, respectively. * denotes methods that require camera intrinsics.

		Per-sequence ATE (m) ↓							Sequence averages		
		Colosseo-0 8815f 1.45 km	Campus-0 12042f 2.73 km	Campus-1 11671f 2.95 km	Pincio-0 11142f 1.27 km	Spagna-0 14141f 1.56 km	Diag-0 10021f 1.02 km	Ciampino-1 18846f 5.20 km	Avg. ATE (m)↓	Avg. RPE$_r$ (°)↓	Avg. RPE$_t$ (m)↓
Group	**Method**										
Opt.-based	VGGT-SLAM	54.30	122.07	117.15	32.48	24.86	33.52	188.30	81.81	1.08	0.31
	MASt3R-SLAM	88.75	116.21	114.37	67.11	54.62	35.67	171.76	92.64	1.25	0.30
	VGGT-Long	44.03	132.67	115.62	64.29	59.04	33.66	179.43	89.82	0.82	0.37
	Droid-SLAM*	98.12	132.85	119.53	63.21	64.20	35.86	198.21	101.71	0.82	0.23
	DPV-SLAM*	86.34	107.85	70.29	51.56	55.60	33.87	105.91	73.06	0.70	0.16
	DPVO*	90.42	109.72	58.49	52.83	57.84	33.84	102.60	72.25	1.34	0.15
	Droid-W*	50.27	125.02	96.89	21.96	22.82	24.59	171.52	73.29	0.58	0.13
Streaming	InfiniteVGGT	89.89	136.76	116.63	77.76	63.35	33.84	202.09	102.90	8.27	4.84
	OVGGT	96.94	137.74	117.39	77.76	62.49	34.84	201.43	104.08	11.43	6.07
	Stream3R-w	92.51	138.38	116.38	77.43	64.57	35.64	202.14	103.87	7.81	2.72
	CUT3R	99.21	136.32	117.92	75.05	62.29	33.66	202.10	103.79	1.50	0.37
	CUT3R w/ reset	76.68	59.51	67.21	48.57	48.55	26.66	169.18	70.91	0.75	0.11
	TTT3R	91.11	129.47	116.59	74.31	60.96	34.39	194.75	100.23	3.58	0.50
	TTT3R w/ reset	73.32	56.51	52.76	37.39	35.33	34.97	175.98	66.61	0.81	0.10
	LongStream	83.01	120.29	106.92	71.39	60.81	30.61	177.51	92.93	0.71	0.13
	LingBot-Map	**11.11**	21.47	12.52	46.80	35.12	15.62	63.26	29.41	4.44	2.69
	HorizonStream	40.89	27.84	24.28	26.77	28.16	25.39	29.83	29.02	**0.60**	**0.04**
	HorizonStream w/ LC	24.66	27.22	8.72	16.29	21.33	**7.36**	**17.92**	17.64	0.61	**0.04**
	Ours	47.93	10.98	17.64	21.02	29.24	28.09	55.96	30.12	**0.60**	**0.04**
	Ours w/ LC	15.82	**8.24**	**5.52**	**10.74**	**13.33**	16.74	24.81	**13.60**	**0.60**	**0.04**

Table 6. Streaming efficiency on KITTI-02. FPS measures inference throughput and peak memory reports GPU occupancy (input excluded).

Method	FPS↑	Mem.↓
InfiniteVGGT	7.30	15.95
OVGGT	11.87	8.05
Stream3R-w	12.76	5.26
CUT3R	29.62	3.16
TTT3R	25.53	4.65
LongStream	10.36	6.62
LongStream*	23.39	7.50
LingBot-Map	19.74	18.87
HorizonStream	8.02	13.04
HorizonStream*	29.14	26.62
Ours	24.45	6.71

Table 7. Dense reconstruction on 7Scenes, TUM-Dynamic, and Oxford Spires. CD is measured in meters (↓), and F1 in percent (↑). Best and second-best results are marked in blue and underlined, respectively.

	7Scenes ($\tau = 0.25\,\mathrm{m}$)		TUM-Dyn. ($\tau = 0.25\,\mathrm{m}$)		Oxford Spires ($\tau = 4\,\mathrm{m}$)	
Method	CD↓	F1↑	CD↓	F1↑	CD↓	F1↑
CUT3R	0.18	75.18	0.11	90.68	7.26	41.22
TTT3R	0.11	86.98	**0.06**	94.45	7.77	40.42
Stream3R-w	0.12	87.57	0.20	87.40	6.44	47.75
InfiniteVGGT	0.08	91.96	0.10	92.94	7.38	47.83
OVGGT	0.07	92.91	0.09	93.91	6.84	55.38
LongStream	0.06	94.60	0.14	88.29	5.69	53.84
LingBot-Map	**0.05**	96.01	0.08	**95.97**	1.68	90.58
HorizonStream	0.23	**98.22**	0.12	90.97	2.00	84.70
Ours	0.06	94.84	0.11	92.21	**1.37**	**91.79**

of composition-aware pose supervision, long-sequence training, and temporal rotation refinement. As shown in Table 8, we start from a baseline trained with sole adjacent-frame pose supervision and progressively introduce each component. Following the evaluation protocol described above, we report ATE, RPE$_r$, and RPE$_t$.

Composition-Aware Pose Supervision. Adding the composition-aware loss functions produces a substantial improvement over the 32-frame baseline on all three datasets. The baseline supervises individual frame-to-frame transformations but does not explicitly constrain the errors produced after these transformations are recursively composed. This creates a mismatch between training, which emphasizes isolated relative poses, and inference, where the complete trajectory is recovered through repeated pose composition. Consequently, small systematic errors can accumulate into considerable trajectory drift. The composition-aware objective is designed to mitigate this issue by explicitly supervising multi-step composed transformations, thereby encouraging relative-pose predictions to maintain consistency over extended temporal intervals. The substantial improvement over the baseline further demonstrates the importance of composition-aware supervision for learning stable trajectories, beyond optimizing the accuracy of individual relative motions.

Long-Sequence Training. Extending the training sequence from 32 to 128 frames further reduces ATE and relative pose errors, while the local pose-prediction window remains unchanged. This change corresponds to the relative improvements of 19.3%, 36.7%, and 17.8%, respectively. We attribute this improvement primarily to the denser, temporally continuous adjacent-pose supervision provided by longer clips. Longer clips also reduce clip-boundary effects and improve the coverage

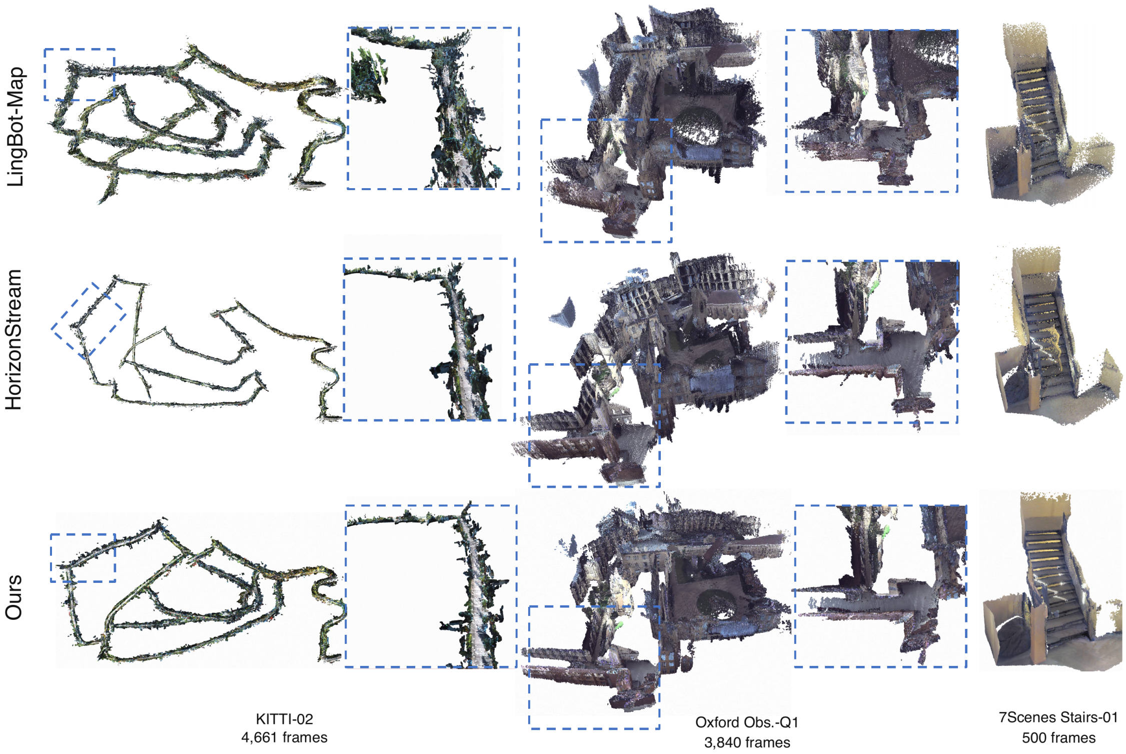


Figure 7. Qualitative dense reconstruction comparison on representative sequences from KITTI, Oxford Spires, and 7Scenes. Dashed boxes indicate regions shown at a larger scale for KITTI-02 and Oxford Obs.-Q1. Across different sequence lengths and scene scales, our method preserves coherent global structure while maintaining well-aligned local geometry using only local temporal context.

Table 8. Ablation study on KITTI, Oxford, and VBR. Starting from the baseline, the evaluated components are introduced incrementally such that consecutive rows differ in only one factor. "Frames" denotes the length of the clips used during training, and "Model size" denotes the number of trainable parameters. Results are averaged over the evaluation sequences of each dataset, and lower values indicate better performance. The best results are highlighted in bold.

	Training configuration				KITTI			Oxford			VBR		
Variant	Frames	Model size	Refin.	$\mathcal{L}_{\text{comp}}$	ATE	RPE$_r$	RPE$_t$	ATE	RPE$_r$	RPE$_t$	ATE	RPE$_r$	RPE$_t$
Baseline	32	930M	–	–	56.60	0.19	0.17	10.16	0.19	0.03	52.90	0.83	0.06
+ Comp. loss	32	930M	–	✓	27.66	0.14	0.13	9.40	0.15	0.03	44.72	0.63	0.06
+ Long-seq. training	128	930M	–	✓	22.31	0.11	0.12	5.95	0.13	**0.02**	36.77	0.62	0.06
+ Rot. refinement	128	935M	✓	✓	**18.30**	**0.10**	**0.10**	**4.35**	**0.12**	**0.02**	**30.12**	**0.60**	**0.04**

of local motion patterns, leading to more consistent pose estimates after composition.

Motion-Visual Contextualized Rotation Refinement. The lightweight temporal rotation refinement module introduces only 4.75M additional parameters (**0.51%** of the full model) while reducing ATE by 18.0%, 26.9%, and 18.1% on KITTI, Oxford Spires, and VBR, respectively. The consistent improvements in both ATE and relative pose errors demonstrate that a small temporal refinement module can effectively suppress local rotational bias and improve long-horizon trajectory composition. The consistent gains across all three datasets demonstrate that temporal refinement remains beneficial even after the model has been trained with long sequences and composition-aware supervision. By incorporating recent motion and appearance context, the refinement module can resolve temporal ambiguities that are difficult to disambiguate from an isolated frame pair.

5 Discussion

Our results show that the proposed local formulation is particularly effective for long, temporally continuous streams. The model maintains stable local pose estimation as the processing horizon grows and remains consistent under longer pose compositions, leading to strong trajectory accuracy on KITTI, VBR, and Oxford Spires. On compact indoor benchmarks, the gains are less pronounced and our reconstruction results are not uniformly the best. One possible reason is that, in spatially confined scenes with frequent revisit and high visual overlap, persistent geometric context can remain useful over long portions of the sequence. Our method does not explicitly retain such scene-level state, yet remains competitive in these settings, suggesting that the local formulation remains effective across both compact and substantially larger environments.

The optional loop-closure backend complements the local predictor when scene revisits are available. Because it is independent of model training, loop closure introduces sparse long-range constraints only at inference time while leaving the learned streaming formulation unchanged. This provides additional global correction on sequences with useful revisits, without requiring persistent learned long-range state inside the predictor. In this sense, local prediction and loop closure play complementary roles: the former provides a bounded, sequence-length-independent streaming backbone, while the latter offers optional global refinement when additional long-range constraints are available.

The resulting modular design also leaves room for further extensions. The local streaming backbone can incorporate advances in feed-forward reconstruction, caching, and inference acceleration without changing the underlying local-to-global formulation. Future work may explore stronger handling of dynamic scenes, larger-scale pretraining, and selective long-range constraints for cases where local temporal continuity becomes ambiguous, as well as applications to long-horizon embodied perception, 3D-vlm and world modeling.

6 Conclusion

We presented *ABot-Recon*, a streaming 3D reconstruction model that formulates long-horizon reconstruction as repeated local prediction and reliable composition. The model predicts current-frame geometry together with adjacent-frame relative pose within a fixed temporal horizon, allowing the learned predictor to operate with bounded state independent of the total sequence length. A lightweight temporal rotation refiner improves local motion estimation, while composition-aware pose supervision explicitly constrains the multi-step transformations used to assemble long trajectories.

Across long driving, handheld, and landmark-scale sequences, *ABot-Recon* achieves robust camera-pose and dense-reconstruction performance while maintaining a compact streaming state. These results show that a well-structured local prediction formulation provides an effective alternative for long-horizon streaming 3D reconstruction without requiring persistent learned long-range state in the predictor. When additional global consistency is desired, an optional training-independent loop-closure backend can further refine the reconstructed trajectory without altering the underlying streaming formulation.

References

[1] S. Wang, V. Leroy, Y. Cabon, B. Chidlovskii, and J. Revaud, “DUSt3R: Geometric 3d vision made easy,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2024, pp. 20 697–20 709.

[2] V. Leroy, Y. Cabon, and J. Revaud, “Grounding image matching in 3d with MASt3R,” in *European Conference on Computer Vision*, ser. Lecture Notes in Computer Science, vol. 15130. Springer, 2024, pp. 71–91.

[3] J. Wang, M. Chen, N. Karaev, A. Vedaldi, C. Rupprecht, and D. Novotny, “VGGT: Visual geometry grounded transformer,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2025, pp. 5294–5306.

[4] Y. Wang, J. Zhou, H. Zhu, W. Chang, Y. Zhou, Z. Li, J. Chen, J. Pang, C. Shen, and T. He, “π^3: Permutation-equivariant visual geometry learning,” in *International Conference on Learning Representations*, 2026.

[5] H. Wang and L. Agapito, “3d reconstruction with spatial memory,” in *International Conference on 3D Vision*, 2025, pp. 78–89.

[6] Q. Wang, Y. Zhang, A. Holynski, A. A. Efros, and A. Kanazawa, “Continuous 3d perception model with persistent state,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2025, pp. 10 510–10 522.

[7] D. Zhuo, W. Zheng, J. Guo, Y. Wu, J. Zhou, and J. Lu, “Streaming visual geometry transformer,” in *International Conference on Learning Representations*, 2026.

[8] Y. Lan, Y. Luo, F. Hong, S. Zhou, H. Chen, Z. Lyu, S. Yang, B. Dai, C. C. Loy, and X. Pan, "STream3R: Scalable sequential 3d reconstruction with causal transformer," in *International Conference on Learning Representations*, 2026.

[9] Z. Li, J. Zhou, Y. Wang, H. Guo, W. Chang, Y. Zhou, H. Zhu, J. Chen, C. Shen, and T. He, "WinT3R: Window-based streaming reconstruction with camera token pool," in *International Conference on Learning Representations*, 2026.

[10] Y. Liu, S. Dong, S. Wang, Y. Yin, Y. Yang, Q. Fan, and B. Chen, "SLAM3R: Real-time dense scene reconstruction from monocular RGB videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2025, pp. 16 651–16 662.

[11] Z. Chen, M. Qin, T. Yuan, Z. Liu, and H. Zhao, "LONG3R: Long sequence streaming 3d reconstruction," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, Oct. 2025, pp. 5273–5284.

[12] C. Cheng, X. Chen, T. Xie, W. Yin, W. Ren, Q. Zhang, X. Guo, and H. Wang, "LongStream: Long-sequence streaming autoregressive visual geometry," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2026, pp. 272–283.

[13] R. Wang, Y. Song, Y. Cai, and L. Liu, "STAC: Plug-and-play spatio-temporal aware cache compression for streaming 3d reconstruction," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2026, pp. 7567–7576.

[14] X. Chen, Y. Chen, Y. Xiu, A. Geiger, and A. Chen, "TTT3R: 3d reconstruction as test-time training," in *International Conference on Learning Representations*, 2026.

[15] C. Liu, J. Yang, Z. Li, Y. Deng, J. Guo, and L. Ballan, "Mem3R: Streaming 3d reconstruction with hybrid memory via test-time training," *arXiv preprint arXiv:2604.07279*, 2026. [Online]. Available: https://arxiv.org/abs/2604.07279

[16] L.-Z. Chen, J. Gao, Y. Chen, K. L. Cheng, Y. Sun, L. Hu, N. Xue, X. Zhu, Y. Shen, Y. Yao, and Y. Xu, "Geometric context transformer for streaming 3d reconstruction," *arXiv preprint arXiv:2604.14141*, 2026. [Online]. Available: https://arxiv.org/abs/2604.14141

[17] C. Cheng, P. Tao, N. Yao, G. Ding, X. Chen, Y. Du, X. Guo, W. Yin, W. Ren, Q. Zhang, Z. Chen, and H. Wang, "HorizonStream: Long-horizon attention for streaming 3d reconstruction," *arXiv preprint arXiv:2605.23889*, 2026. [Online]. Available: https://arxiv.org/abs/2605.23889

[18] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? the KITTI vision benchmark suite," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2012, pp. 3354–3361.

[19] Y. Tao, M. Á. Muñoz-Bañón, L. Zhang, J. Wang, L. F. T. Fu, and M. Fallon, "The oxford spires dataset: Benchmarking large-scale LiDAR-visual localisation, reconstruction and radiance field methods," *The International Journal of Robotics Research*, vol. 45, no. 6, pp. 839–857, 2026.

[20] J. L. Schönberger and J.-M. Frahm, "Structure-from-motion revisited," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2016, pp. 4104–4113.

[21] C. Campos, R. Elvira, J. J. Gómez Rodríguez, J. M. M. Montiel, and J. D. Tardós, "ORB-SLAM3: An accurate open-source library for visual, visual–inertial, and multimap SLAM," *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.

[22] Z. Tang, Y. Fan, D. Wang, H. Xu, R. Ranjan, A. G. Schwing, and Z. Yan, "MV-DUSt3R+: Single-stage scene reconstruction from sparse views in 2 seconds," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2025, pp. 5283–5293.

[23] J. Yang, A. Sax, K. J. Liang, M. Henaff, H. Tang, A. Cao, J. Chai, F. Meier, and M. Feiszli, "Fast3R: Towards 3d reconstruction of 1000+ images in one forward pass," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2025, pp. 21 924–21 935.

[24] S. Zhang, J. Wang, Y. Xu, N. Xue, C. Rupprecht, X. Zhou, Y. Shen, and G. Wetzstein, "FLARE: Feed-forward geometry, appearance and camera estimation from uncalibrated sparse views," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2025, pp. 21 936–21 947.

[25] C. Liu, B. Tan, Z. Ke, S. Zhang, J. Liu, M. Qian, N. Xue, Y. Shen, and T. Braud, "PLANA3R: Zero-shot metric planar 3d reconstruction via feed-forward planar splatting," in *Advances in Neural Information Processing Systems*, 2025.

[26] J. Zhang, C. Herrmann, J. Hur, V. Jampani, T. Darrell, F. Cole, D. Sun, and M.-H. Yang, “MonST3R: A simple approach for estimating geometry in the presence of motion,” in *International Conference on Learning Representations*, 2025.

[27] Y. Wu, W. Zheng, J. Zhou, and J. Lu, “Point3R: Streaming 3d reconstruction with explicit spatial pointer memory,” in *Advances in Neural Information Processing Systems*, 2025.

[28] Z. Teed and J. Deng, “DROID-SLAM: Deep visual SLAM for monocular, stereo, and RGB-D cameras,” in *Advances in Neural Information Processing Systems*, 2021, pp. 16 558–16 569.

[29] R. Murai, E. Dexheimer, and A. J. Davison, “MASt3R-SLAM: Real-time dense SLAM with 3d reconstruction priors,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2025, pp. 16 695–16 705.

[30] K. Deng, Z. Ti, J. Xu, J. Yang, and J. Xie, “VGGT-Long: Chunk it, loop it, align it—pushing VGGT’s limits on kilometer-scale long RGB sequences,” *arXiv preprint arXiv:2507.16443*, 2025. [Online]. Available: https://arxiv.org/abs/2507.16443

[31] F. Zhang, T. Zhang, K. Khosoussi, Z. Zhang, Z. Huang, and Y. Luo, “TALO: Pushing 3d vision foundation models towards globally consistent online reconstruction,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2026, pp. 21 870–21 879.

[32] P. Tao, C. Cheng, Y. Du, C. Song, Z. Chen, X. Guo, W. Yin, W. Ren, Q. Zhang, H. Cui, and S. Shen, “Anchor3R: Streaming 3d reconstruction with transient anchors for long-horizon visual mapping,” *arXiv preprint arXiv:2606.05035*, 2026. [Online]. Available: https://arxiv.org/abs/2606.05035

[33] C. Xu, H. Gao, X. Chen, Y. Xiu, J. Gao, and A. Chen, “R^3: 3d reconstruction via relative regression,” *arXiv preprint arXiv:2605.26519*, 2026. [Online]. Available: https://arxiv.org/abs/2605.26519

[34] L. Mou, R. Men, G. Li, Y. Xu, L. Zhang, R. Yan, and Z. Jin, “Natural language inference by tree-based convolution and heuristic matching,” in *Annual Meeting of the Association for Computational Linguistics*, 2016, pp. 130–136.

[35] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” *Advances in neural information processing systems*, vol. 30, 2017.

[36] C. Lea, M. D. Flynn, R. Vidal, A. Reiter, and G. D. Hager, “Temporal convolutional networks for action segmentation and detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, 2017, pp. 1003–1012. [Online]. Available: https://arxiv.org/pdf/1611.05267

[37] W. Wang, D. Zhu, X. Wang, Y. Hu, Y. Qiu, C. Wang, Y. Hu, A. Kapoor, and S. Scherer, “Tartanair: A dataset to push the limits of visual slam,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2020, pp. 4909–4916.

[38] L. Ling, Y. Sheng, Z. Tu, W. Zhao, C. Xin, K. Wan, L. Yu, Q. Guo, Z. Yu, Y. Lu *et al.*, “Dl3dv-10k: A large-scale scene dataset for deep learning-based 3d vision,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 22 160–22 169.

[39] P. Sun, H. Kretzschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine *et al.*, “Scalability in perception for autonomous driving: Waymo open dataset,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 2446–2454.

[40] Y. Zhou, Y. Wang, J. Zhou, W. Chang, H. Guo, Z. Li, K. Ma, X. Li, Y. Wang, H. Zhu *et al.*, “Omniworld: A multi-domain and multi-modal dataset for 4d world modeling,” in *International Conference on Learning Representations*, 2026.

[41] G. Baruch, Z. Chen, A. Dehghan, T. Dimry, Y. Feigin, P. Fu, T. Gebauer, B. Joffe, D. Kurz, A. Schwartz *et al.*, “Arkitscenes: A diverse real-world dataset for 3d indoor scene understanding using mobile rgb-d data,” *arXiv preprint arXiv:2111.08897*, 2021.

[42] M. Patel, F. Yang, Y. Qiu, C. Cadena, S. Scherer, M. Hutter, and W. Wang, “Tartanground: A large-scale dataset for ground robot perception and navigation,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025.

[43] C. Yeshwanth, Y.-C. Liu, M. Nießner, and A. Dai, “Scannet++: A high-fidelity dataset of 3d indoor scenes,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 12–22.

[44] Y. Cabon, N. Murray, and M. Humenberger, "Virtual kitti 2," *arXiv preprint arXiv:2001.10773*, 2020.

[45] H. Xia, Y. Fu, S. Liu, and X. Wang, "Rgbd objects in the wild: Scaling real-world 3d object learning from rgb-d videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 22 378–22 389.

[46] M. Roberts, J. Ramapuram, A. Ranjan, A. Kumar, M. A. Bautista, N. Paczan, R. Webb, and J. M. Susskind, "Hypersim: A photorealistic synthetic dataset for holistic indoor scene understanding," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10 912–10 922.

[47] A. Dai, A. X. Chang, M. Savva, M. Halber, T. Funkhouser, and M. Nießner, "Scannet: Richly-annotated 3d reconstructions of indoor scenes," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2017, pp. 5828–5839.

[48] Y. Zheng, A. W. Harley, B. Shen, G. Wetzstein, and L. J. Guibas, "Pointodyssey: A large-scale synthetic dataset for long-term point tracking," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 19 855–19 865.

[49] E. Arnold, J. Wynn, S. Vicente, G. Garcia-Hernando, Á. Monszpart, V. A. Prisacariu, D. Turmukhambetov, and E. Brachmann, "Map-free visual relocalization: Metric pose relative to a single image," in *European Conference on Computer Vision*, 2022.

[50] F. Tosi, Y. Liao, C. Schmitt, and A. Geiger, "SMD-Nets: Stereo mixture density networks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 8942–8952.

[51] G. Baruch, Z. Chen, A. Dehghan, T. Dimry, Y. Feigin, P. Fu, T. Gebauer, B. Joffe, D. Kurz, A. Schwartz *et al.*, "Arkitscenes: A diverse real-world dataset for 3d indoor scene understanding using mobile rgb-d data," *arXiv preprint arXiv:2111.08897*, 2021.

[52] L. Mehl, J. Schmalfuss, A. Jahedi, Y. Nalivayko, and A. Bruhn, "Spring: A high-resolution high-detail dataset and benchmark for scene flow, optical flow and stereo," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 4981–4991.

[53] Z. Bauer, F. Gomez-Donoso, E. Cruz, S. Orts-Escolano, and M. Cazorla, "Uasol, a large-scale high-resolution outdoor stereo dataset," *Scientific data*, vol. 6, no. 1, p. 162, 2019.

[54] P.-H. Huang, K. Matzen, J. Kopf, N. Ahuja, and J.-B. Huang, "Deepmvs: Learning multi-view stereopsis," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2018, pp. 2821–2830.

[55] V. Guizilini, R. Ambrus, S. Pillai, A. Raventos, and A. Gaidon, "3d packing for self-supervised monocular depth estimation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020.

[56] N. Karaev, I. Rocco, B. Graham, N. Neverova, A. Vedaldi, and C. Rupprecht, "Dynamicstereo: Consistent dynamic depth from stereo videos," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 13 229–13 239.

[57] Y. Liao, J. Xie, and A. Geiger, "KITTI-360: A novel dataset and benchmarks for urban scene understanding in 2d and 3d," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 3, pp. 3292–3310, 2023.

[58] A. Avetisyan *et al.*, "Scenescript: Reconstructing scenes with an autoregressive structured language model," in *European Conference on Computer Vision*, 2024.

[59] J. Ou, Z. Cao, Y. Ren, Z. Li, J. Zhu, T. Hua, S. Zhang, H. Xiong, and W. Zhao, "Holo360d: A large-scale real-world dataset with continuous trajectories for advancing panoramic 3d reconstruction and beyond," in *European Conference on Computer Vision*, 2026. [Online]. Available: https://arxiv.org/abs/2604.22482

[60] M. Fonder and M. Van Droogenbroeck, "Mid-air: A multi-modal dataset for extremely low altitude drone flights," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2019, pp. 553–562.

[61] Y. Li, L. Jiang, L. Xu, Y. Xiangli, Z. Wang, D. Lin, and B. Dai, "Matrixcity: A large-scale city dataset for city-scale neural rendering and beyond," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[62] X. Cheng, R. Wu, X. Liu, Z. Cui, Y. Liu, N. Zhao, Y. Liu, M. Zhang, and S. Yan, "Airzoo: A unified large-scale dataset for grounding aerial geometric 3d vision," 2026. [Online]. Available: https://arxiv.org/abs/2604.26567

[63] Y. Yao, Z. Luo, S. Li, J. Zhang, Y. Ren, L. Zhou, T. Fang, and L. Quan, "Blendedmvs: A large-scale dataset for generalized multi-view stereo networks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 1790–1799.

[64] L. Brizi, E. Giacomini, L. Di Giammarino, S. Ferrari, O. Salem, L. De Rebotti, and G. Grisetti, "VBR: A vision benchmark in rome," in *IEEE International Conference on Robotics and Automation*, 2024.

[65] J. Shotton, B. Glocker, C. Zach, S. Izadi, A. Criminisi, and A. Fitzgibbon, "Scene coordinate regression forests for camera relocalization in rgb-d images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2013, pp. 2930–2937.

[66] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, "A benchmark for the evaluation of rgb-d slam systems," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*. IEEE, 2012, pp. 573–580.

[67] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations*, 2019.

[68] B. T. Polyak and A. B. Juditsky, "Acceleration of stochastic approximation by averaging," *SIAM Journal on Control and Optimization*, vol. 30, no. 4, pp. 838–855, 1992.

[69] Z. Ye, L. Chen, R. Lai, W. Lin, Y. Zhang, S. Wang, T. Chen, B. Kasikci, V. Grover, A. Krishnamurthy, and L. Ceze, "Flashinfer: Efficient and customizable attention engine for llm inference serving," in *Proceedings of Machine Learning and Systems*, 2025. [Online]. Available: https://arxiv.org/abs/2501.01005

[70] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with gpus," *IEEE transactions on big data*, vol. 7, no. 3, pp. 535–547, 2019.

[71] S. Izquierdo and J. Civera, "Optimal transport aggregation for visual place recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, 2024, pp. 17 658–17 668.

[72] S. Yuan, Y. Yang, X. Yang, X. Zhang, Z. Zhao, L. Zhang, and Z. Zhang, "Infinitevggt: Visual geometry grounded transformer for endless streams," 2026.

[73] S.-Y. Lu, P.-T. Chen, H.-C. Hsu, S.-Y. Jhong, W.-H. Cheng, and Y.-Y. Chen, "Ovggt: O (1) constant-cost streaming visual geometry transformer," *arXiv preprint arXiv:2603.05959*, 2026.

[74] D. Maggio, H. Lim, and L. Carlone, "VGGT-SLAM: Dense RGB SLAM optimized on the SL(4) manifold," in *Advances in Neural Information Processing Systems*, 2025.

[75] Z. Fang, W. Zheng, Y. Yuan, W. Wang, Z. Chen, C. Sun, J. Huang, K. Li, M. Qin, and H. Zhao, "Slamformer-∞: Infinite slam transformer for unbounded frontend and backend processing," 2026. [Online]. Available: https://arxiv.org/abs/2608.03429

[76] L. Lipson, Z. Teed, and J. Deng, "Deep patch visual slam," in *European Conference on Computer Vision*, 2024.

[77] Z. Teed, L. Lipson, and J. Deng, "Deep patch visual odometry," in *Advances in Neural Information Processing Systems*, 2023.

[78] M. Li, Z. Zhu, M. Pollefeys, and D. Barath, "Droid-slam in the wild," 2026. [Online]. Available: https://arxiv.org/abs/2603.19076

[79] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *International journal of computer vision*, vol. 60, no. 2, pp. 91–110, 2004.

A Data Processing

A.1 Temporal Sampling

Video datasets. We construct every training sample as an ordered sequence. For video datasets, the interval $[a, b]$ denotes the range of frame-index steps between consecutive inputs. We use three temporal sampling strategies according to the structure and length of the source trajectory.

1. **Forward fixed-stride sampling.** We uniformly sample one feasible interval from $[a, b]$ and use it throughout the clip. Frames are always traversed in chronological order. This strategy is used for long trajectories that can provide the requested number of frames without reversing direction.

2. **Adaptive forward sampling.** We sample only from the forward portion of a trajectory. When the remaining sequence is too short for the configured maximum interval, we reduce the upper bound to fit the complete clip. Depending on the dataset, the resulting clip uses either one fixed stride or independently sampled per-step strides.

3. **Foldback sampling.** We traverse the source sequence with a stride sampled log-uniformly from $[a, b]$. Upon reaching a sequence boundary, the traversal reverses direction and a new stride is sampled for the next segment. Foldback sampling allows a short source video to provide a long training clip while preserving local continuity and avoiding jumps between unrelated frames.

Multi-view datasets. For unordered multi-view data, we construct fixed-length sequences using a camera pose graph. Let θ_{ij} denote the relative rotation between views i and j, and let $d_{ij} = \|\mathbf{t}_i - \mathbf{t}_j\|_2 / s$ denote their translation normalized by the median nearest-neighbor camera distance s. For each view, we consider its K nearest neighbors and retain at most K_g edges satisfying $\theta_{ij} < \theta_{\max}$ and $d_{ij} < \tau_t$. Beam search then expands valid paths and keeps the best B candidates according to their motion and turning costs. Repeated views are permitted only through valid graph edges, subject to a minimum unique-view ratio $\rho_{\min}$.

Table 9. Pose-graph sequence sampling parameters.

Dataset	$\theta_{\max}$	τ_t	K	K_g	B	$\rho_{\min}$
HyperSim	20°	5	48	24	192	0.50
US4K	20°	5	64	24	128	1.00
BlendedMVS	40°	5	24	14	96	0.75

US4K requires all views to be unique, while HyperSim and BlendedMVS allow constrained revisiting when necessary.

Datasets percentage. The temporal sampling strategy or multi-view construction rule used for each dataset is summarized in Table 10.

A.2 Data Filtering

We find that public datasets contain occasional annotation errors and noisy samples. By inspecting sampled training examples and analyzing the samples associated with loss spikes during training, we identify problematic data and apply dataset-specific filtering. Removing these erroneous samples substantially improves model performance.

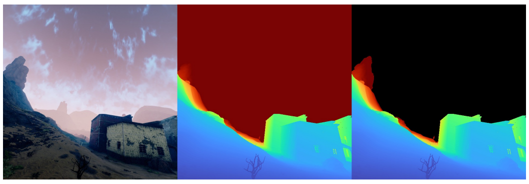
BlendedMVS. We observed that some BlendedMVS scenes contain sideways or upside-down images, as shown in Fig. 8. We manually identified and excluded 38 such scenes.

Figure 8. Examples of sideways and upside-down images in BlendedMVS.

TartanAir & TartanAir v2. For the TartanAir datasets, we found that some scenes contained erroneous depth values in sky regions, as shown in Fig. 9(a). We invalidated the corresponding depth values using semantic masks. We also observed erroneous water-surface depths in the Ocean scene, as shown in Fig. 9(b), and therefore excluded this scene from both TartanAir and TartanAir-v2.

Table 10. Per-dataset sequence-construction and interval policies used for training, resolved from the training configuration and dataset-loader defaults. Interval bounds are measured in source frames. For pose-graph data, b_{med} denotes the scene's median nearest-neighbor camera baseline.

Dataset	Type	Sampling strategy	Interval / constraint
TartanAir-v2	Synth.	Foldback	Hard $[1, 8]$, Easy $[1, 16]$
TartanAir	Synth.	Adaptive forward	$[1, 20]$
OmniWorld-Game	Synth.	Foldback	$[4, 16]$
TartanGround	Synth.	Foldback	$[1, 8]$
Virtual KITTI 2	Synth.	Adaptive forward	$[1, 5]$
HyperSim	Synth.	Pose-graph path	$\Delta R < 20°$, $\Delta t < 5b_{\mathrm{med}}$
PointOdyssey	Synth.	Adaptive forward	$[1, 4]$
Unreal4K	Synth.	Pose-graph path	$\Delta R < 20°$, $\Delta t < 5b_{\mathrm{med}}$
Spring	Synth.	Foldback	$[1, 4]$
MVS-Synth	Synth.	Foldback	$[1, 4]$
Dynamic Replica	Synth.	Adaptive forward	$[1, 16]$
ASE	Synth.	Foldback	$[1, 2]$
MidAir	Synth.	Foldback	$[1, 8]$
MatrixCity	Synth.	Forward fixed-stride	1
AirZoo	Synth.	Foldback	$[1, 8]$
BlendedMVS	Synth.	Pose-graph path	$\Delta R < 40°$, $\Delta t < 5b_{\mathrm{med}}$
Internal Synthetic	Synth.	Forward fixed-stride	$[2, 4]$
DL3DV	Real	Adaptive forward	$[1, 20]$
Waymo	Real	Adaptive forward	$[1, 8]$
ARKitScenes	Real	Foldback	1
ScanNet++	Real	Foldback	1
WildRGBD	Real	Foldback	$[1, 4]$
ScanNet	Real	Adaptive forward	$[1, 30]$
MapFree	Real	Adaptive forward	$[1, 30]$
ARKitScenes HR	Real	Foldback	1
UASOL	Real	Adaptive forward	$[1, 40]$
DDAD	Real	Forward fixed-stride	1
KITTI-360	Real	Forward fixed-stride	$[2, 6]$
Internal Real	Real	Forward fixed-stride	$[1, 2]$
Holo360D	Real	Foldback	$[1, 2]$
Holo360	Real	Foldback	$[1, 2]$


RGB Origin Depth Masked Depth
(a) Incorrect Depth

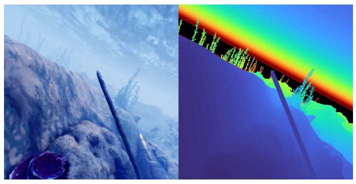

RGB Depth
(b) Ocean

Figure 9. Examples of incorrect scenes in TartanAir and TartanAir-v2.

OmniWorld-Game. We observed erroneous geometric annotations in the dataset, particularly anomalous focal lengths in the camera intrinsics, as shown in Fig. 10. We therefore remove unreliable scenes by validating cross-frame correspondences. Specifically, we sample multiple image pairs at different temporal positions in each sequence and obtain reliable 2D correspondences using SIFT [79] feature matching and fundamental-matrix RANSAC. The matched pixels are then back-projected into 3D using the annotated depths and camera intrinsics. We estimate the relative rigid motion between the two frames and compare it with the annotated camera pose. Rotation error, translation direction and scale errors, and depth-normalized geometric residuals are jointly used to identify inconsistent frame pairs. A sequence is removed when severe inconsistencies account for at least 15% of its valid frame pairs and form a cluster of at least three pairs.


RGB Point Cloud RGB Point Cloud

Figure 10. Examples of incorrect scenes in OmniWorld-Game.

DL3DV. Since DL3DV is constructed from captured videos using structure-from-motion (SfM) and multi-view stereo (MVS), some scenes contain erroneous pose reconstructions. We assume that scenes with discontinuous camera trajectories are likely to suffer from reconstruction failures. Therefore, we remove any scene in which the rotation between two consecutive frames exceeds 70°, or the inter-frame translation exceeds $15\times$ median inter-frame translation of that scene. Moreover, because the DL3DV depth maps contain noise, we first discard the farthest 2% of depth values, apply a morphological opening with a 5×5 kernel, and then perform erosion with a 3×3 kernel to remove isolated depth outliers.

HyperSim. We found that all sampled RGB frames from `ai_003_001/cam_00` and `ai_004_009/cam_01` were completely black; therefore, we excluded both sequences from training.

Scannet++. We observed floating artifacts within the reconstructed meshes of some scenes, which occluded large portions of otherwise valid depth. We therefore excluded the affected scenes from training.

ARKitScenes. We found that the low-resolution depth maps in ARKitScenes, captured using the iPhone LiDAR sensor, are not sufficiently accurate. Therefore, we do not use these depth maps for supervision.

A.3 Spatial Preprocessing and Augmentation

With probability 0.9, we preserve the source field of view by resizing the image isotropically to 504 pixels in width and producing a 504×280 canvas through vertical center cropping or symmetric mean-color padding. The padding value is $(0.485, 0.456, 0.406)$ in normalized RGB space. Geometric annotations and camera intrinsics are transformed consistently with the image. The alternative branch applies principal-point-centered crop and focal-length augmentation. Holo360D and Holo360 additionally retain their original off-center principal points with probability 0.9. Photometric transformations are applied only to valid image content; synthetic padding is restored to its mean value afterward. The photometric pipeline comprises color jitter (brightness, contrast, saturation, hue, and gamma), JPEG degradation, and image blur. For 20% of training sequences, a single set of augmentation parameters—including the JPEG and blur decisions—is sampled and shared by all frames. For the remaining 80%, these parameters are sampled independently for each frame. This mixed strategy retains temporal coherence in part of the training data while exposing the model to frame-wise photometric variation.

B Runtime and Memory Evaluation Protocol

We evaluate runtime and memory efficiency on the full 4,661-frame KITTI-02 sequence at stride 1 using a single NVIDIA H100 GPU. Each method uses its specified inference precision and input resolution. In addition to frame-by-frame streaming, we report the released chunk-based execution modes of LongStream and HorizonStream, denoted by an asterisk. Before measurement, we perform an untimed 64-frame warm-up and then reset the streaming state. The complete 4,661-frame sequence is subsequently processed for timing. All input frames are preloaded onto the GPU, and image loading and preprocessing are excluded. We synchronize CUDA once immediately before and after the complete forward pass; thus, the reported FPS includes model inference and all streaming-state updates, without introducing per-frame synchronization overhead. Peak memory is reported as the maximum PyTorch-allocated GPU memory excluding the storage occupied by the preloaded input sequence. It includes model parameters, intermediate activations, cached tensors, and persistent streaming state. All methods are evaluated on the same sequence and hardware using the same timing and memory accounting protocol. The resulting throughput and memory measurements are reported in Table 6.

C More Results

We further evaluate the generalization of our method on long videos collected outside the benchmarks in the main paper. These sequences are not associated with accurate ground-truth camera trajectories; therefore, the results in this section are intended as qualitative evidence rather than quantitative comparisons. We consider two complementary settings: low-viewpoint robot navigation and large-scale real-world traversal.

Low-viewpoint robot navigation. Figure 11 presents representative results on videos captured by a quadruped robot. This setting differs noticeably from our training data. The camera is mounted close to the ground, causing the ground plane to occupy a large portion of the image while reducing the visibility of distant scene structures, making frame-to-frame correspondence more challenging

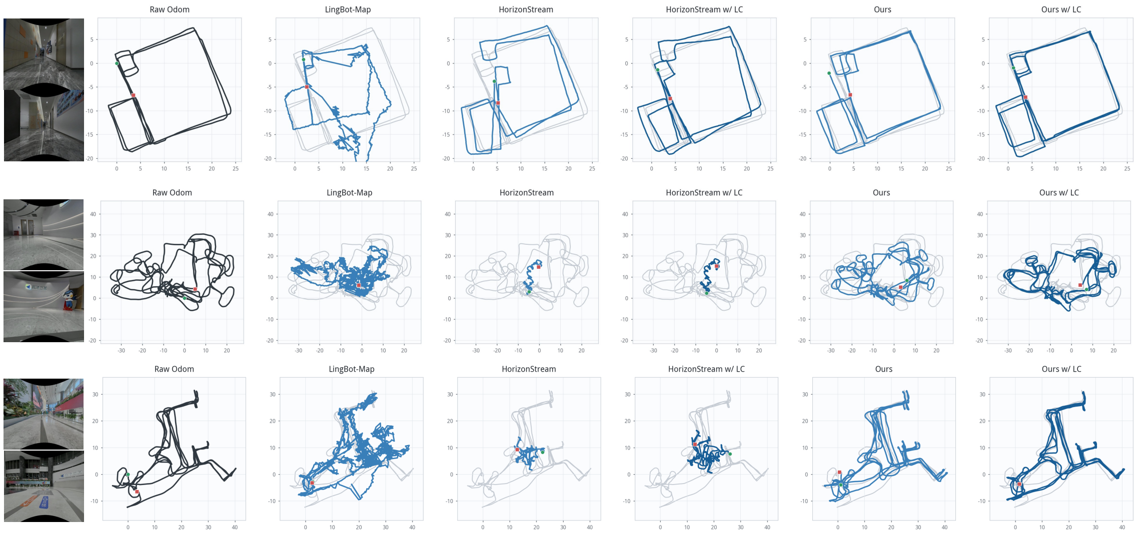


Figure 11. Qualitative camera trajectories on representative quadruped-robot sequences. The low camera height results in ground-dominant observations and motion patterns that differ from the training distribution.

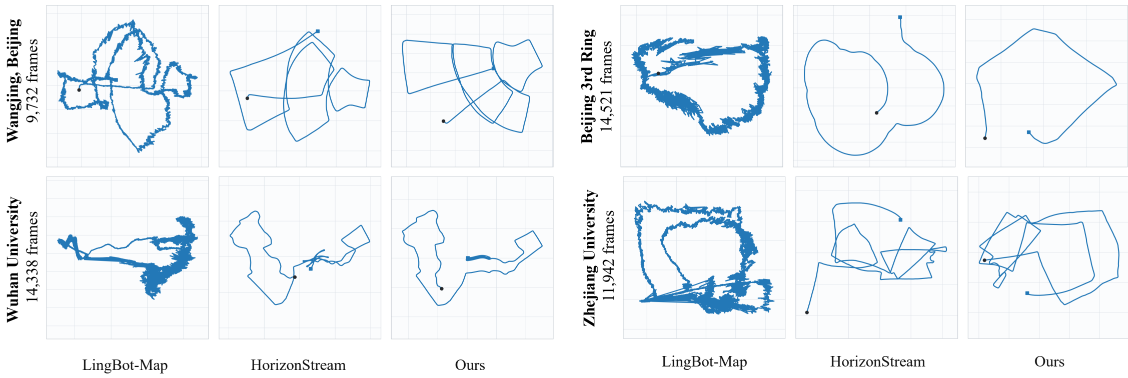


Figure 12. Qualitative no-loop trajectory comparisons on four large-scale real-world videos. The results illustrate the behavior of streaming models under variations in scene scale, camera motion, and data distribution.

than in conventional handheld or vehicle-mounted videos.

Despite this domain shift, our method produces more coherent trajectories across robot-navigation sequences than other methods. These examples provide qualitative evidence that our method can maintain stable trajectory estimates under camera viewpoints and motion patterns that are less frequently represented in the training data.

Large-scale real-world sequences. We additionally evaluate on four long videos captured in the Beijing Third Ring Road, Wuhan University, Zhejiang University and Wangjing area, as shown in Figure 12. These sequences cover different scene scales and capture distributions, including extended urban driving, campus roads, repeated structures, intersections, and substantial variations in camera motion and visual appearance. They therefore complement the standard benchmarks with less curated and more heterogeneous long-video streams.

Across the three sequences, our method produces trajectories with more consistent large-scale structure than the compared streaming baselines. This behavior is consistent with the motivation of our design: the network always predicts an adjacent relative transform, so its regression target does not change as the explored area or elapsed sequence length increases. The lightweight temporal correction further suppresses locally correlated rotational errors before they accumulate through pose composition.

Taken together, these qualitative results extend the benchmark evaluation to camera viewpoints, motion patterns, and spatial scales. While accurate ground truth is unavailable for quantitative assessment, the observed trajectory consistency suggests that keeping pose prediction local provides a practical basis for generalization across heterogeneous long-video streams.